MULTI-AGENT ELECTION LAB

Who wins depends on who you hear

A simulated town of 300 voters. We change one condition at a time and watch who wins.

Team Eclipse · 9 November 2026

THE QUESTION

Does the winner depend only on what people prefer, or also on how they reason and what they can see?

EXPERIMENT 1

Everyone sees everything

Full information · 30 towns · 40 rounds

EXPERIMENT 2

How voters reason

84 voter mixes × 30 towns

EXPERIMENT 3

Who they can hear

5 hearing radii, walking or still, × 30 towns

EXPERIMENT 4

Who tries to sway them

A loyal 10% with 0 to 80 followers × 30 towns

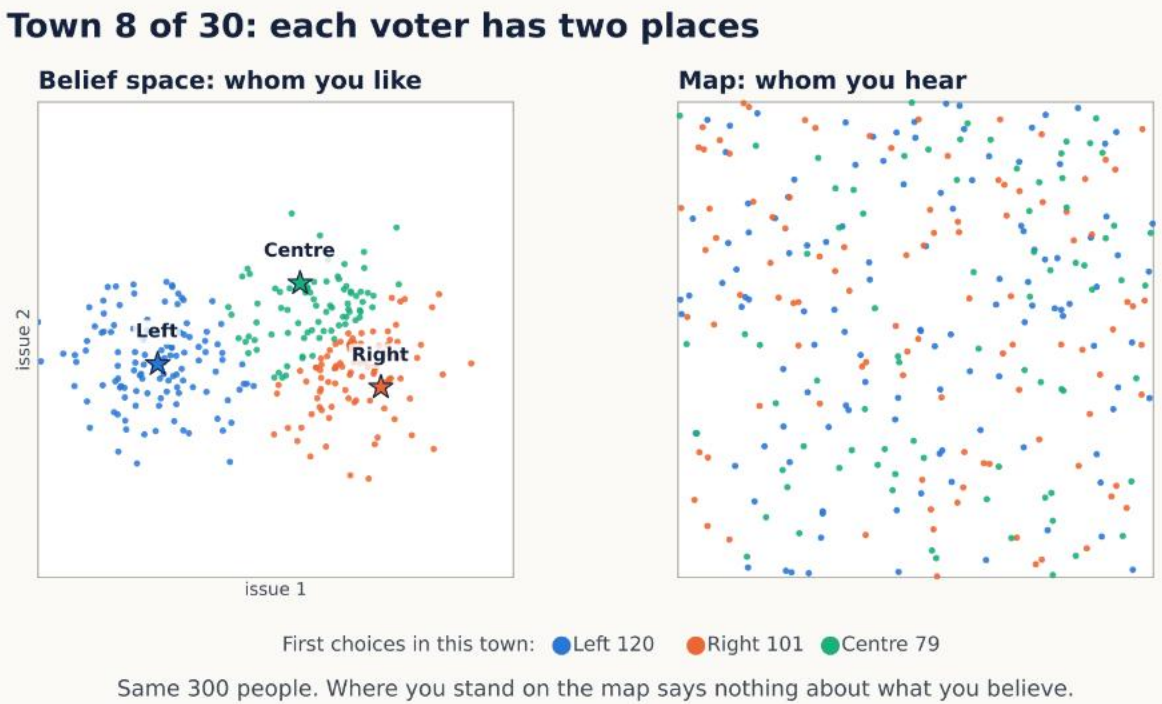
EXPERIMENT 5

Who catches them

A watchdog that sees only vote totals · 10 attack styles

THE TOWN

300 voters, each in two places



Town 8 of 30. First choices: Left 120, Right 101, Centre 79. Each dot keeps its colour, its favourite, as it moves to a random spot on the map.

BELIEF SPACE: WHOM YOU LIKE

A fixed point on two issues. You rank Left, Right and Centre by distance, nearest first.

TWO CAMPS

40% near Left. 60% on the right side, split between Right and Centre.

MAP: WHOM YOU HEAR

A random spot, unrelated to your beliefs. It matters from Experiment 3.

Opinions: 120 voters drawn around (0.25, 0.45), 180 around (0.64, 0.50), spread 0.10 per issue · same recipe in all 30 towns

THE TOWN

The counting rules disagree

118.5
first choices for Left

98
for Right

83
for Centre

HEAD TO HEAD
Centre beats Left **180 to 120** and Right **198 to 102**.

Counting rule	Winner
Plurality: most first choices	**Left**
Instant runoff	**Right** (27 of 30)
Borda points	**Centre**
Beats everyone head to head	**Centre**

From here on: **plurality, one name per voter**.

Every result is over **30 towns**: 30 random draws of this recipe, and the same 30 in every experiment.

Medians over the 30 towns; ranges: Left 111–128, Right 87–110, Centre 70–99 · Centre wins both head-to-heads in all 30

VOTER RULES

The 4 Types of Voters

Sincere · 40%

“I vote for my favourite.”

Control

Front-runner · 30%

“Only the top two can win. I stay with mine if it's there, else I pick the better of the two.”

Tsang et al. 2018, Top-K

Copycat · 20%

“I vote for the most common vote I see.”

Local majority

Pivot-seeker · 10%

“I switch only if my one vote changes the winner.”

Baumeister et al. 2020

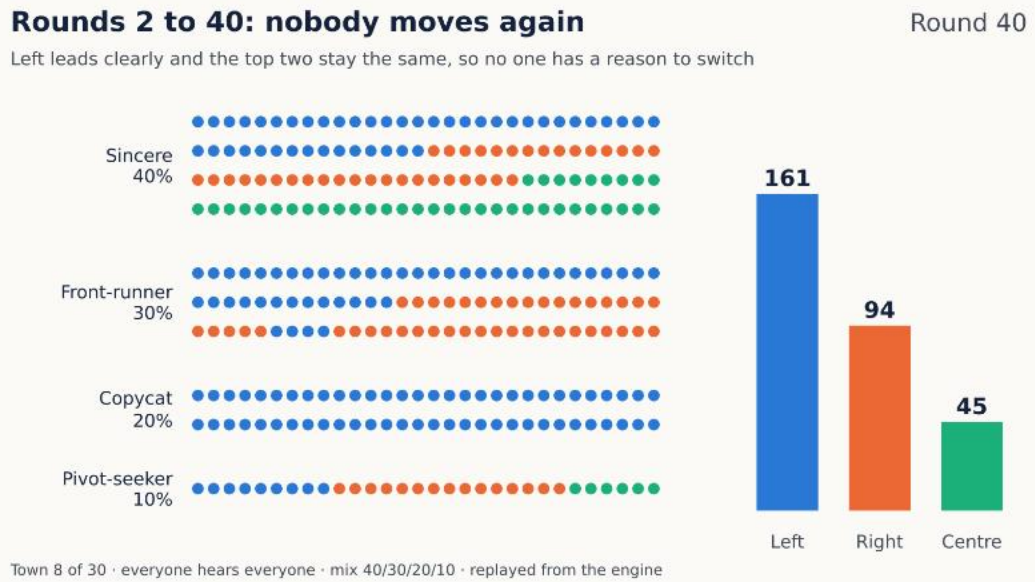
Round 0: everyone votes for their favourite.

Every round after: everyone votes at once, reacting to last round's votes they can see.

Engine policies: sincere, top_k (K = 2), sensed_plurality, pivotal · types are assigned at random, whatever a voter's opinion

EXPERIMENT 1 · THE BASELINE

Everyone hears everyone



Town 8 of 30. Round 0, everyone votes their favourite: Left 120, Right 101, Centre 79. In round 1, 62 voters switch: 161, 94, 45. Then nobody moves.

THE BASELINE

Everyone sees the true count of last round's votes. Default mix 40/30/20/10.

WHAT WE GOT

Frozen after one round in all 30 towns. Left wins all 30; Centre falls from 83 to 41 votes (median).

WHY IT STOPS

Round 1's count has the same top two as round 0, so it points everyone to the vote they already cast. Proved for each of the 30 towns.

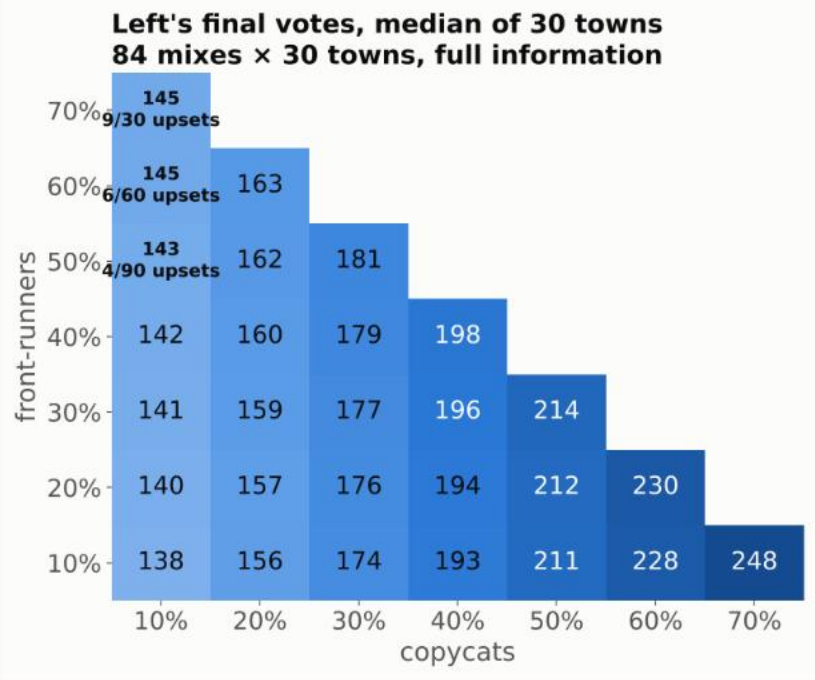
WHY 30 TOWNS

Each town is one random draw of opinions, map spots and voter types. One town could be luck; 30 show the pattern. The same 30 are used in every experiment.

Medians over the 30 towns: Left 118.5 → 159 votes, Right 98 → 98.5, Centre 83 → 41

EXPERIMENT 2

The voter mix



Up: share of front-runners. Across: share of copycats. The rest are sincere voters and pivot-seekers. Each number: Left's final votes, median of 30 towns.

WHAT WE CHANGED

The shares of the four types: all 84 mixes in 10% steps, each type at least 10%. Still full information.

WHAT WE EXPECTED

Different mixes give different winners.

WHAT WE GOT

Left wins 2,501 of 2,520 elections; the 19 upsets go 10 to Right and 9 to Centre. Each extra 10% of copycats adds about 18 votes to Left. Pivot-seekers act like sincere voters.

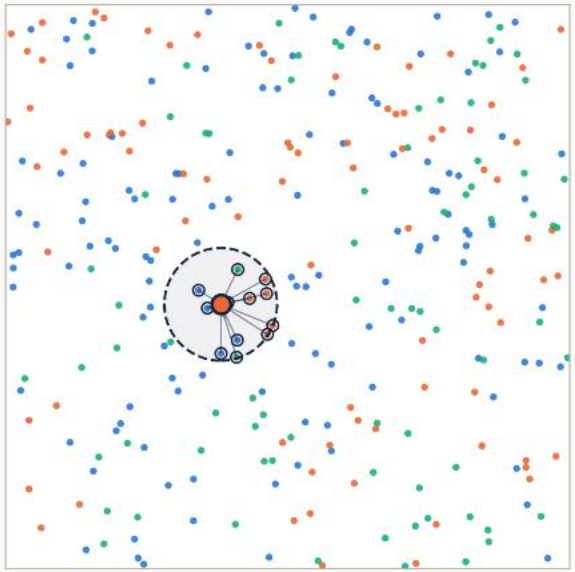
OUR REAL-WORLD READING

With shared information, whether a society has more herders or more tacticians may mostly decide the size of the landslide, not who wins.

Medians: Left 138 → 248 as copycats go 10% → 70% · Centre 66.5 → 16 as strategic voters grow · upsets need 50%+ front-runners

SETUP FOR EXPERIMENT 3

Who you hear: a circle that moves with you



One copycat's hearing circle

Town 8 of 30, radius 0.1, people walking.
This copycat ranks Right > Centre > Left.

Round 3

It hears 12 people's votes from round 2:



Most common: Right
→ it votes Right

Everyone else follows their own rule at the same time.
Replayed from the engine.

Town 8 of 30, radius 0.1, walking, rounds 0 to 12. This copycat ranks Right > Centre > Left, hears 7 to 13 people a round and switches 5 times. Shown: round 3, it hears 12 (Left 4, Right 6, Centre 2) and votes Right.

HEARING RADIUS

You only see last round's votes of people inside your circle. Its size is fixed for a run, and it moves with you.

WALKING

Each round everyone takes a small random step, so the people you hear keep changing.

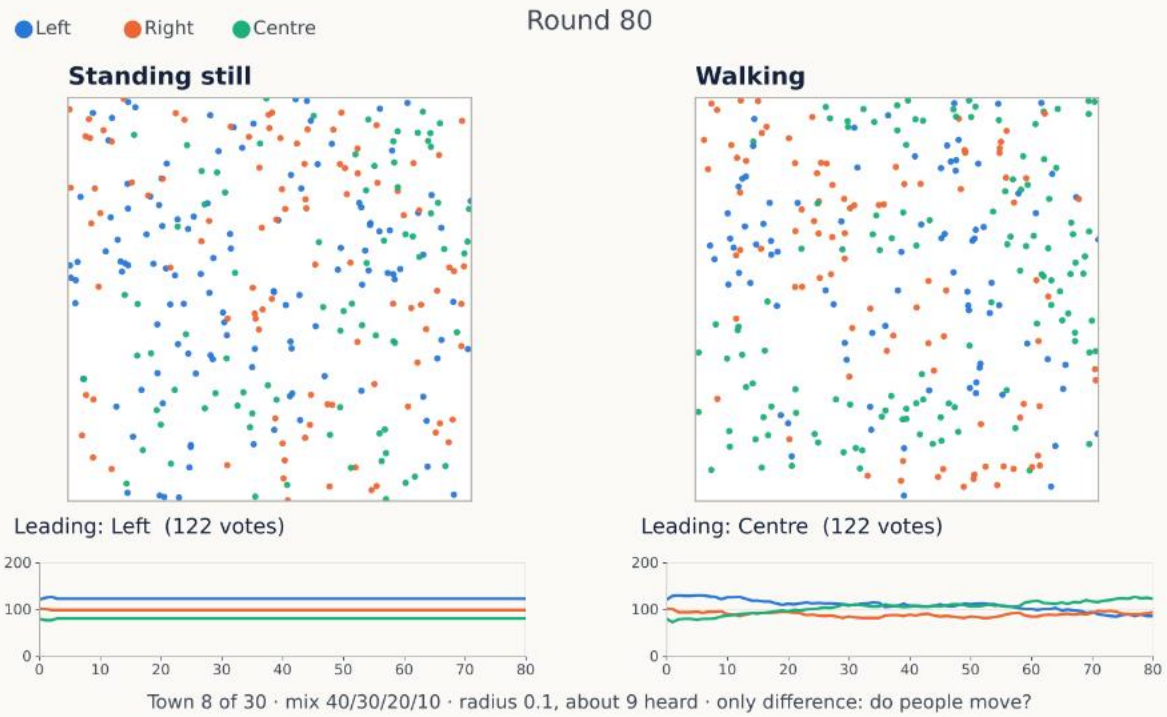
Beliefs stay fixed. Only contacts change.

Radius	People heard
0.4 · 0.2 · 0.1	104 · 32 · 9
0.07 · 0.05	4 · 2

Clip replayed from the engine · each round everyone takes a random step of at most 0.03 on the 1 × 1 map

EXPERIMENT 3 · ONE TOWN

One town, standing still vs walking



Town 8 of 30, mix 40/30/20/10, radius 0.1 (about 9 heard), 80 rounds. The only difference between the two maps: whether people move.

Standing still: Left leads every round. Round 80: Left 122, Right 98, Centre 80. About 6 voters flip back and forth each round.

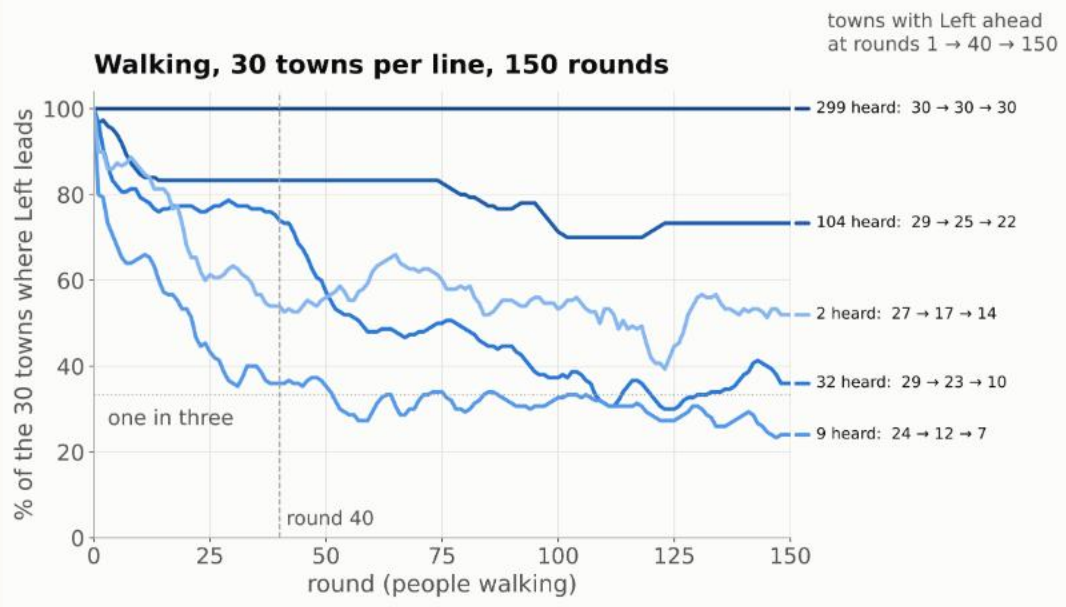
Walking: Left and Centre trade the lead 11 times (rounds 31 to 57). Round 80: Centre 122, Right 93, Left 85. About 11 voters switch each round.

Same people, same opinions. Only whether contacts change.

Both runs replayed from the engine · walking: a random step of at most 0.03 per round on the 1 × 1 map

EXPERIMENT 3

Hearing only neighbours



How to read it: each line is one hearing size, with everyone walking. Its height is the share of the 30 towns where Left, first in round 0 in every town, is still ahead. Right edge: how many towns at rounds 1, 40 and 150.

WHAT WE CHANGED

How many people you hear, from about 104 down to about 2. Walking or standing still.

WHAT WE GOT

Standing still, towns settle; Left wins 23 to 26 of 30. Walking with 32 heard or fewer, votes never settle. With 32 heard, Left leads in 29 of 30 towns after round 1, 10 by round 150.

WHY

Your circle is a small sample, and nearby Right or Centre may be ahead. Copycats and front-runners react to that local picture, so Left's town-wide bandwagon never fully forms. Walking changes the sample every round, so people keep re-deciding.

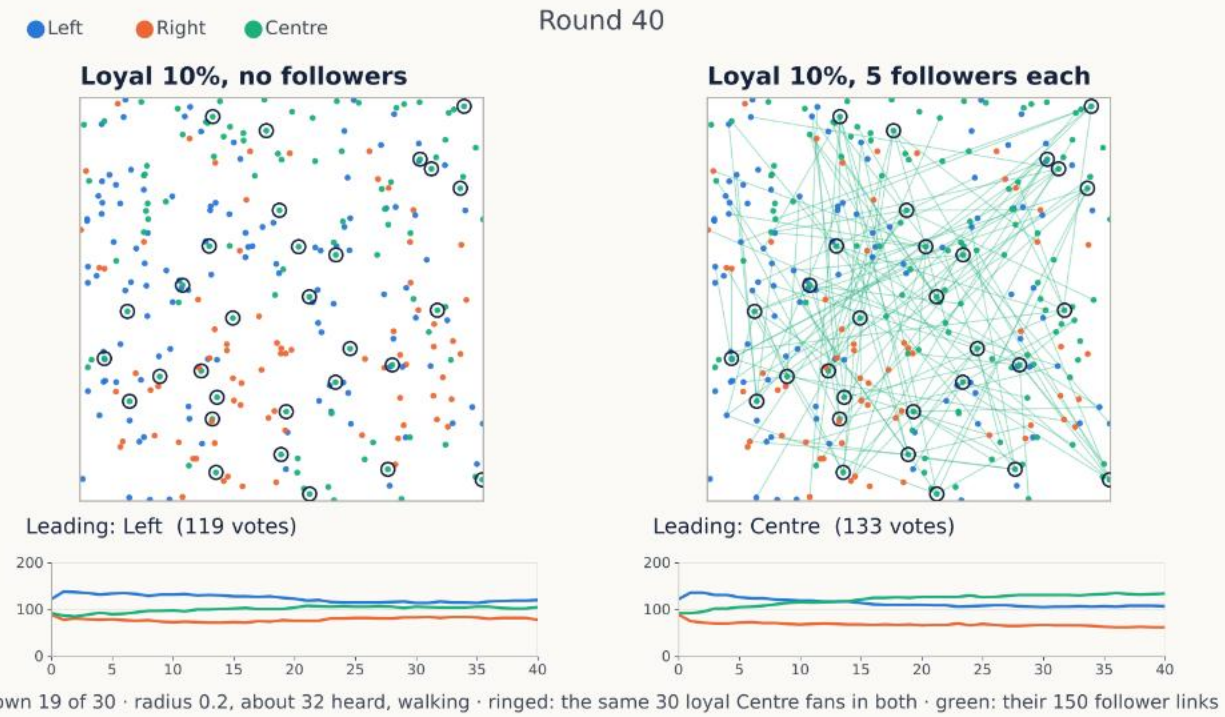
OUR REAL-WORLD READING

Each round is an election every four years: if circles reshuffle, the same voters with the same beliefs can elect different winners.

Engine detail: front-runners are sticky with 300 voters; with 301, votes still never settle and Left wins 20–28 of 30

SETUP FOR EXPERIMENT 4

An organised minority with an audience



Town 19 of 30, radius 0.2, walking, 40 rounds. Both maps have the same 30 loyal Centre fans, ringed. Only on the right do they have followers: 5 each, the green lines.

30 Centre fans, 10% of the town, always vote Centre.

Each has **5 followers**: random voters anywhere in town who see that member's vote every round, like a social media audience.

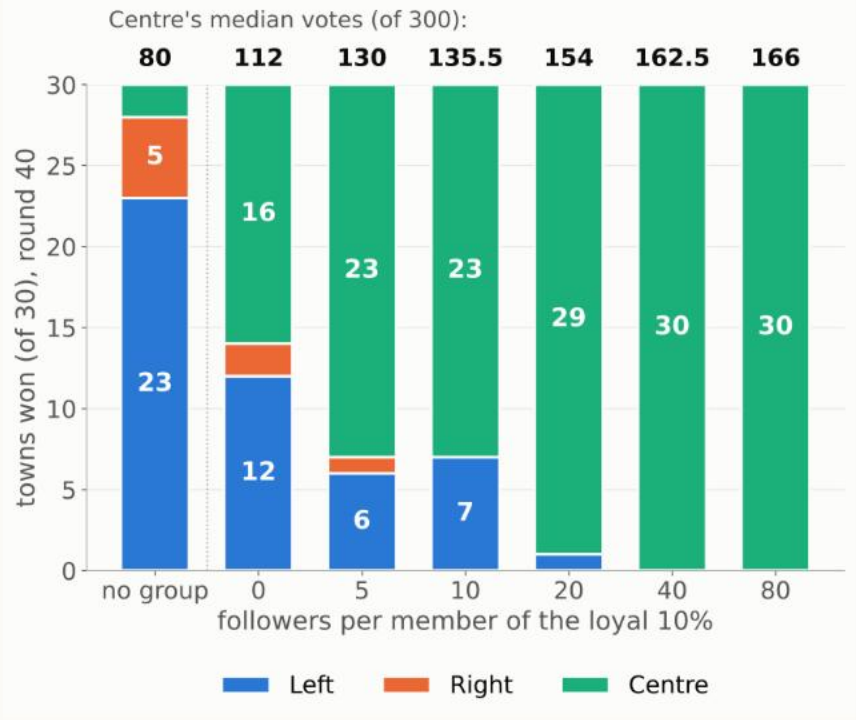
150 links: about **2 in 5 voters** also see a loyal fan's vote.

Everyone else hears about 32 neighbours and walks.

Round 40 in town 19 · no followers: Left 119, Centre 104, Right 77 · 5 followers each: Centre 133, Left 106, Right 61

EXPERIMENT 4

Propaganda



30 towns per bar, round 40, radius 0.2, walking. On top: Centre's median votes. First bar: no group.

WHAT WE CHANGED

First loyalty: 30 Centre fans who never waver. Then reach: 0 to 80 followers per member.

WHAT WE GOT

Loyalty alone: Centre wins 16 of 30 towns, up from 2. Just 5 followers each: 23. With 20: 29, and it holds over 150 rounds.

WHY IT WORKS

Centre is almost everyone's second choice: wherever it looks like one of the top two nearby, tactical voters switch to it, and where it leads, copycats follow. Loyal fans keep it in that top two; one extra Centre vote, seen by a follower, can tip a close local count.

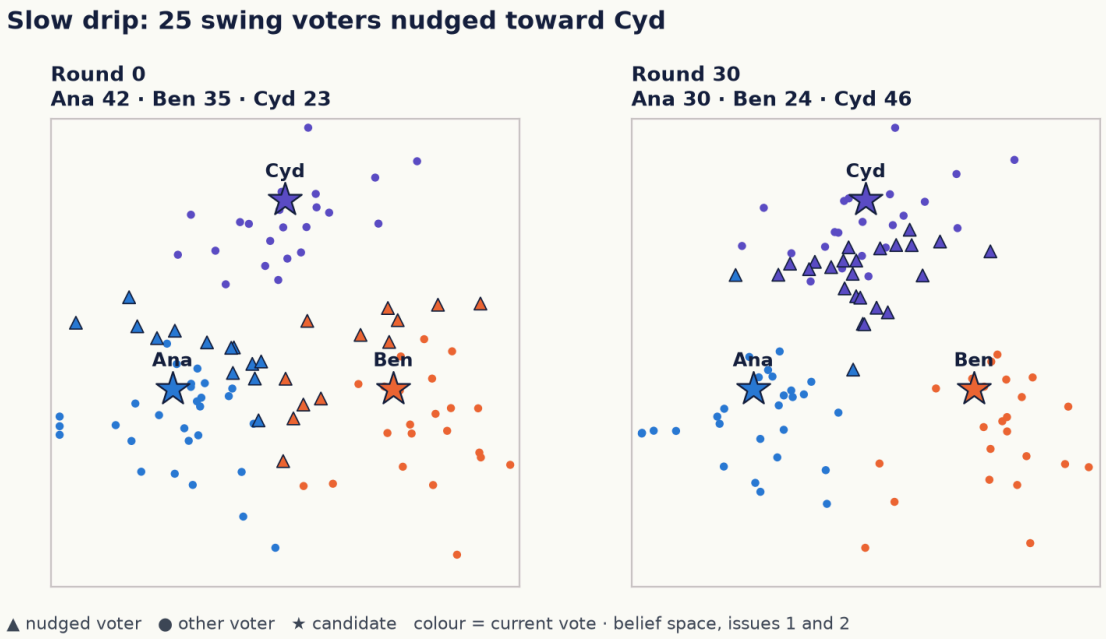
OUR REAL-WORLD READING

Being heard, not hearing, is the lever: a small group with an audience could swing an election without changing anyone's beliefs.

Radius 0.2, about 32 heard, walking, round 40 · with 301 voters: 13, 15, 24 and 30 of 30 for 0, 5, 20 and 40 followers

SETUP FOR EXPERIMENT 5

Can a watchdog catch a slow takeover?



One attack, start and end. Triangles are the voters being pushed. Over 30 rounds they drift toward Cyd, and Cyd goes from third (23 votes) to first (46).

THE TOWN

100 people, three candidates: **Ana 42, Ben 35, Cyd 23**. Everyone votes for the candidate closest to their views.

THE ATTACKER

Has a budget to nudge chosen voters' views toward Cyd, a little every round. We tried 10 styles: fast or slow, few or many voters.

THE WATCHDOG

Sees **only the vote totals**, like an election monitor. It wins if it sounds the alarm **before Cyd wins three rounds in a row**.

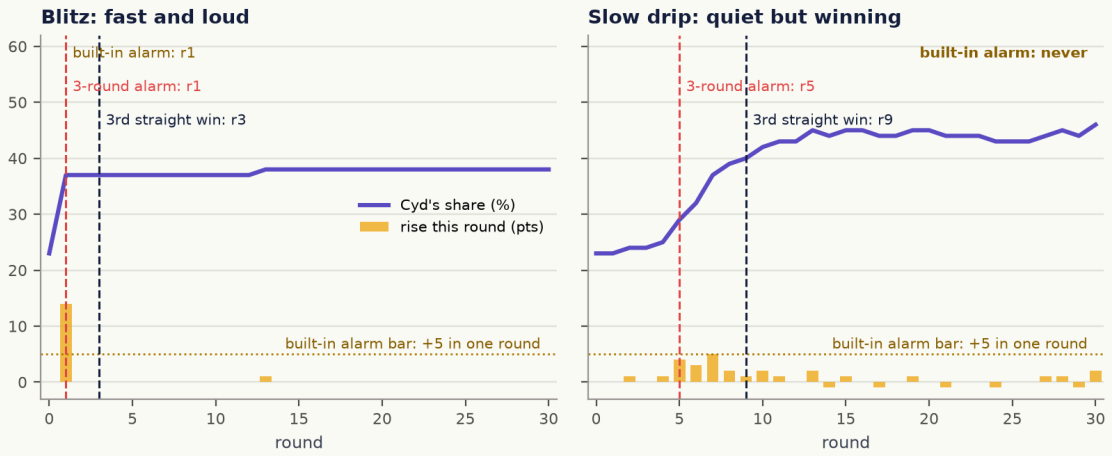
THE BUILT-IN ALARM

Fires if the target jumps more than 5 points in a single round.

Views also drift a little on their own every round · 30 rounds · every attack has an untouched twin town

EXPERIMENT 5

Watch the trend, not the jump



Purple: Cyd's vote share. **Yellow:** how much it rose that round. The blitz jumps at once; the slow drip never rises more than 5 in a round, yet wins by round 9.

WHAT WE GOT

The one-round alarm caught **20 of 25** winning attacks: it misses slow, patient ones. Our best rule, **"alarm if the target rose more than 4 points over 3 rounds"**, caught **25 of 25**, with **no false alarms** in 40 untouched towns.

CAN THE ATTACKER DODGE IT?

We let the attacker try 36 plans. Against the one-round alarm, a slow drip wins every time unseen. Against the 3-round rule, **none** got through.

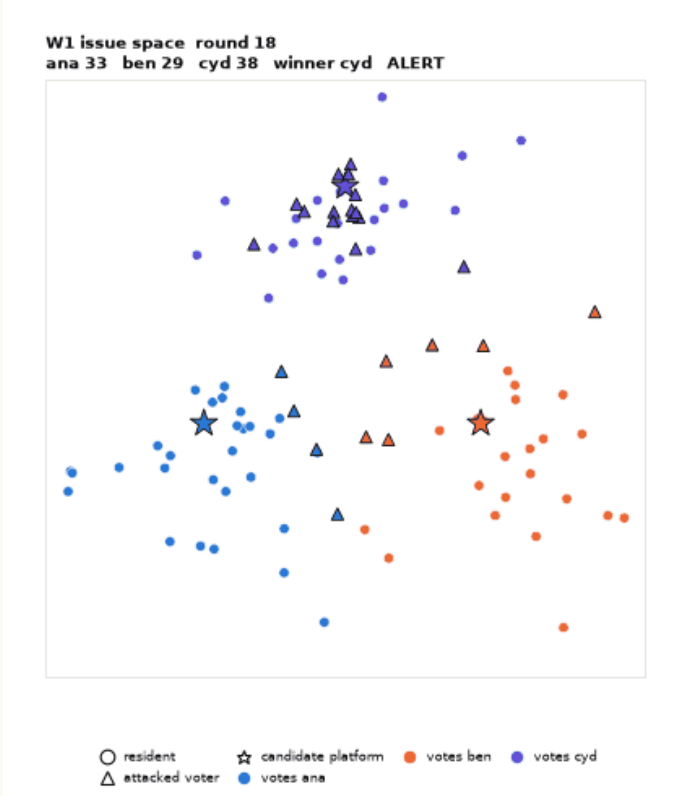
OUR REAL-WORLD READING

Monitors that only look for sudden swings miss patient manipulation. Looking a few rounds back is enough to catch it.

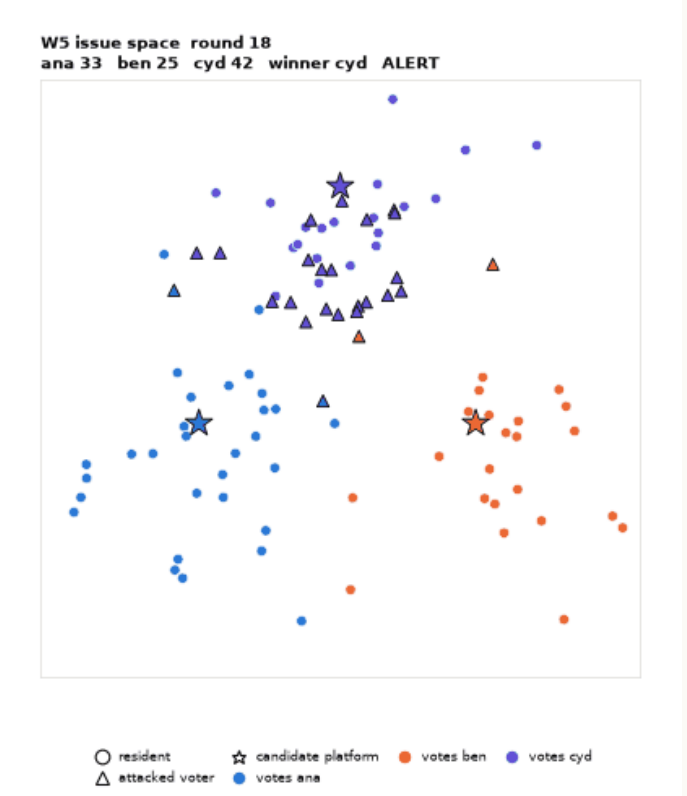
Best rule picked from 101 candidates on one set of towns, then tested on new ones · a 3-point one-round bar also works

EXPERIMENT 5 · WATCH IT HAPPEN

A blitz is loud, a slow drip is quiet



Blitz: Cyd jumps 14 points in round 1. Every alarm fires at once.



Slow drip: Cyd creeps up a few points a round and still takes the lead.

Belief space · ▲ pushed voters · ★ candidates · colour = current vote · 'ALERT' appears when the 3-round alarm fires

TAKEAWAYS

1. With shared information the first leader locks in: in all 30 towns after one round, proved for each.

2. The mix of voters moves the margin, almost never the winner: 19 upsets in 2,520 elections.

3. Changing contacts make the winner unstable, even with fixed beliefs: with 32 heard, Left leads in 29 of 30 towns after round 1, 10 by round 150.

4. A loyal minority with even a small audience can decide the result: with 5 followers each, Centre goes from 2 to 23 of 30 towns.

5. A watchdog must watch the trend, not the jump: a 3-round look-back caught 25 of 25 slow takeovers, with no false alarms.

BEYOND THIS TALK

This talk is one slice of our project

The simulator behind it does much more. All of this is built and tested; the details are in our project report.

ALSO BUILT IN OUR SIMULATOR

Voters that provably settle

One voter moves at a time, with convergence proofs for best response, local dominance and local conformity.

Coalitions

Private channels, false reports, eavesdropping and, on a budget, a look at others' preferences.

Polls and knowledge

Public polls that can be distorted, and voters that reason over possible worlds about what others know.

Campaigns

A campaigner with a budget who buys new links to chosen voters.

Leaders and opinions

Candidates nominated from the town, leaders who adapt, attacks that shift opinions, and a detector for them.

Ten strategic voter rules

From expected utility to maximin. The exact ones blow up with more parties, so they suit small races.

NEXT EXPERIMENTS

Mixed voter types (each a blend, say 60% sincere and 40% front-runner) · unequal hearing: hubs and echo chambers · ranked ballots and more candidates · realistic opinions that drift

APPENDIX

Everything behind the results

MATHEMATICAL MODEL

The town as equations

Preferences, hearing, the four decision rules, the parity effect and the full dynamics, in five parts

PROOFS AND THEORY

Why it locks in, why it wanders

The one-step lock-in proof, small-sample odds and stability, and the memory-based voter

EXTRA EXPERIMENT

Campaigns vs echo chambers

Three parties, five campaign playbooks, all 125 combinations

Mathematical slides adapted from Rishi's version · extra experiment run on a separate town of 100 people

APPENDIX · MATHEMATICAL FORMULATION

The model in one slide

Preferences	$n = 300$, $C = \{\text{Left, Right, Centre}\}$: $c \succ_i c' \Leftrightarrow \|\theta_i - \kappa(c)\| < \|\theta_i - \kappa(c')\|$
Hearing set	$H_i(t) = \{\, j \neq i : \|x_i(t) - x_j(t)\| \leq r \,\}$, plus follower links in Experiment 4
Observation	$o_i(t) = \{\, b_j(t-1) : j \in H_i(t) \,\}$: last round's ballots only
Policy	$b_i(t) = \sigma_i(o_i(t))$, $\sigma_i \in \{\text{sincere, copycat, front-runner, pivot-seeker}\}$
Outcome	$s(c, t) = \|\{\, i : b_i(t) = c \,\}\|$, winner $W(t) = \arg\max_c s(c, t)$
Process	$X(t+1) = \Phi(\, X(t), \xi(t) \,)$: a Markov chain whose only randomness is walking

The next five slides spell out each part

Voters, preferences and the plurality rule

Agents, candidates	$N = \{1, \ldots, n\}$, $n = 300$ · $C = \{\text{Left, Right, Centre}\}$, $m = 3$
Preferences	Ideal point θ_i, platform $\kappa(c)$ in $[0,1]^2$: $c \succ_i c' \Leftrightarrow \lVert\theta_i - \kappa(c)\rVert < \lVert\theta_i - \kappa(c')\rVert$
Utility	$u_i(c) = (m - 1 - \text{rank}_i(c)) \,/\, (m - 1) \in \{1, ½, 0\}$; rank 0 = favourite
Ballot, tally	$b_i(t) \in C$, $s(c, t) = \lvert\{\, i : b_i(t) = c \,\}\rvert$, $b_i(0) = \text{top}_i$
Winner	$W(s) = \arg\max_c s(c)$; ties broken by a fixed party order
Town generator	$\theta \sim \text{Normal}((0.25, 0.45), 0.1^2 I)$ for 120 voters, $\text{Normal}((0.64, 0.50), 0.1^2 I)$ for 180; $x_i(0) \sim \text{Uniform}[0,1]^2$; random type · 30 independent towns

Information: who hears whom

Walking	$x_i(t+1) = \text{reflect}(\, x_i(t) + \ell \cdot (\cos\varphi, \sin\varphi)\,)$, $\varphi \sim U[0, 2\pi)$, $\ell \sim U[0, 0.03]$; standing still: no step
Hearing graph	$j \to i$ in $G(t) \Leftrightarrow j \neq i$ and $\lVert x_i(t) - x_j(t) \rVert \leq r$; full information = complete graph
Followers (Exp. 4)	Fixed edges $F \subset \Lambda \times N$: each loyal voter $\lambda \in \Lambda$ gets f uniformly random followers
Hearing set	$H_i(t) = \{\, j : j \to i \text{ in } G(t) \,\} \cup \{\, \lambda : (\lambda, i) \in F \,\}$
Observation	$o_i(t) = \{\, (j, b_j(t-1)) : j \in H_i(t) \,\}$, $n_i(c, t) = \lvert\{\, j \in H_i(t) : b_j(t-1) = c \,\}\rvert$

Hearing size ≈ $(n - 1)\pi r^2$: 9.4 at $r = 0.1$ (measured ≈ 9) · map edges cut big circles: 150 predicted at $r = 0.4$, 104 measured

Decision rules: each agent's policy

Sincere 40% · control	$b_i(t) = \text{top}_i$, the first choice in $\succ_i$, every round
Copycat 20% · local majority	$b_i(t) = \arg\max_c n_i(c, t)$, ties to the earlier candidate in $\succ_i$; sincere if $H_i(t) = \emptyset$
Front-runner 30% · Top-2, next slide	Restrict to the two apparent leaders $T_i(t)$; weigh each by the chance that a tie between them makes your vote decisive
Pivot-seeker 10% · pivotal voter	Belief $\hat{s}_i$ = round-0 tally, updated with every observed switch. Outcome $w_i(c) = W(\hat{s}_i - e(b_i) + e(c))$. Switch to $c^* = \arg\max u_i(w_i(c))$ only if $u_i(w_i(c^*)) > u_i(w_i(b_i))$

Engine policies: sincere, sensed_plurality, top_k (K = 2), pivotal · e(c) = unit vector for candidate c · types assigned independently of opinion

The front-runner rule and a parity effect

1 · Smoothed local estimate	$\hat{p}_i(c) = (n_i(c) + 1) / (\lvert H_i \rvert + m)$
2 · Apparent leaders	$T_i = \{x, y\}$, the two largest $\hat{p}_i$; $q = \hat{p}_i(x) / (\hat{p}_i(x) + \hat{p}_i(y))$
3 · Chance your vote decides	$\pi(x, y) = P[\ \text{Binomial}(n - 1, q) = (n - 1)/2\ ]$
4 · Score and choice	$V_i(x) = \pi(x, y) \cdot (u_i(x) - u_i(y))$; $V_i = -\infty$ outside T_i. Keep b_i if it attains max V_i, else the best by $\succ_i$

Why 300 and 301 voters differ

$n = 300$: 299 others, an odd number, so an exact tie is impossible and $\pi = 0$.

Inertia decides: stay if already on a leader, else take your preferred leader. A **sticky** front-runner.

$n = 301$: $\pi > 0$, so you always take your preferred leader.

Dynamics: one round and the process

ONE ROUND, T → T + 1

1. Move: new positions x(t+1)
2. Rebuild G(t+1) and hearing sets
3. Observe last round's ballots b(t)
4. All decide at once: $b_i(t+1) = \sigma_i(o_i)$
5. Tally s(t+1), winner W

State	$X(t) = (x(t), b(t), \hat{s}(t))$: positions, ballots, pivot beliefs
Transition	$X(t+1) = \Phi(X(t), \xi(t))$, ξ = walking noise: a Markov chain; deterministic if standing still
Game view	$\langle N, A_i = C, o_i, \Phi, u_i(W) \rangle$ with fixed strategies σ_i
Measured over towns k = 1… 30	$\text{wins}(c) = \Sigma_k \, 1[W(s_k(T)) = c]$; switch rate $\rho(t) = \lvert\{ i : b_i(t) \neq b_i(t-1) \}\rvert / n$

APPENDIX · EXPERIMENT 1 · WHY IT STOPS

One-step lock-in under full information

Proposition

Full information, $n = 300$. If $s(1)$ has the same leader and the same top two as $s(0)$, with gaps of at least 2 so that no single ballot changes either, then $b(2) = b(1)$, and so $b(t) = b(1)$ for all $t \geq 1$.

Checked in all 30 towns: round 1 keeps round 0's top two (Left–Right in 27, Left–Centre in 3).

PROOF, TYPE BY TYPE

Sincere	Ballot never changes.
Copycat	Copies the leader; same leader, same vote.
Front-runner	With $\pi = 0$ it depends only on T; its round-1 vote is already in T, so it stays.
Pivot-seeker	Gap ≥ 2: no vote is decisive, so it stays.

Same ballots give the same observations, so by induction the profile is a fixed point.

Not automatic: with near-ties the same rules can cycle forever, which is why the condition is checked town by town

APPENDIX · EXPERIMENT 3 · WHY

Why: small samples, moving samples

Local samples

P(Left is the strict leader among k heard), round-0 shares:

k	2	9	32	104
P	16%	41%	52%	63%

Most copycats see another leader, so Left's bandwagon never forms.

Standing still

$X(t+1) = \Phi(X(t))$ on a finite set: 3^n ballot profiles × bounded beliefs.

So every run is eventually periodic, guaranteed. We observe period 1 or 2, as for majority rules on symmetric graphs (Goles & Olivos 1980).

Walking

Random contacts: no absorbing profile need exist.

On a finite grid the chain is finite, so the share of rounds each candidate leads converges almost surely (ergodic theorem). The stable object is a frequency, not a winner.

Probabilities: multinomial with median round-0 shares 118.5 / 98 / 83 of 300 · finite-grid result: the engine's stochastic-stability track

APPENDIX · ONGOING ENGINE WORK

A voter with memory and a fading poll

Recent memory, K = 10	$q_i(c)$ = share of c among the latest ballots of everyone met in the last K rounds
Poll every L = 10 rounds	fading weight $w(t) = (L - d + 1)/L$, $d = 1 + ((t-1) \bmod L)$: 1, 0.9, ..., 0.1
Blended estimate	$\hat{p}_i(c) = w \cdot P(c)/\Sigma P + (1-w) \cdot q_i(c)$
Decision	$F(a) = W((n-1)\hat{p}_i + e(a))$; vote for your favourite reachable winner c* only if $c^* \succ_i F(b_i)$

Convergence

A 4-voter cycle refutes the original version; no guarantee for n ≥ 100.

Finite grid

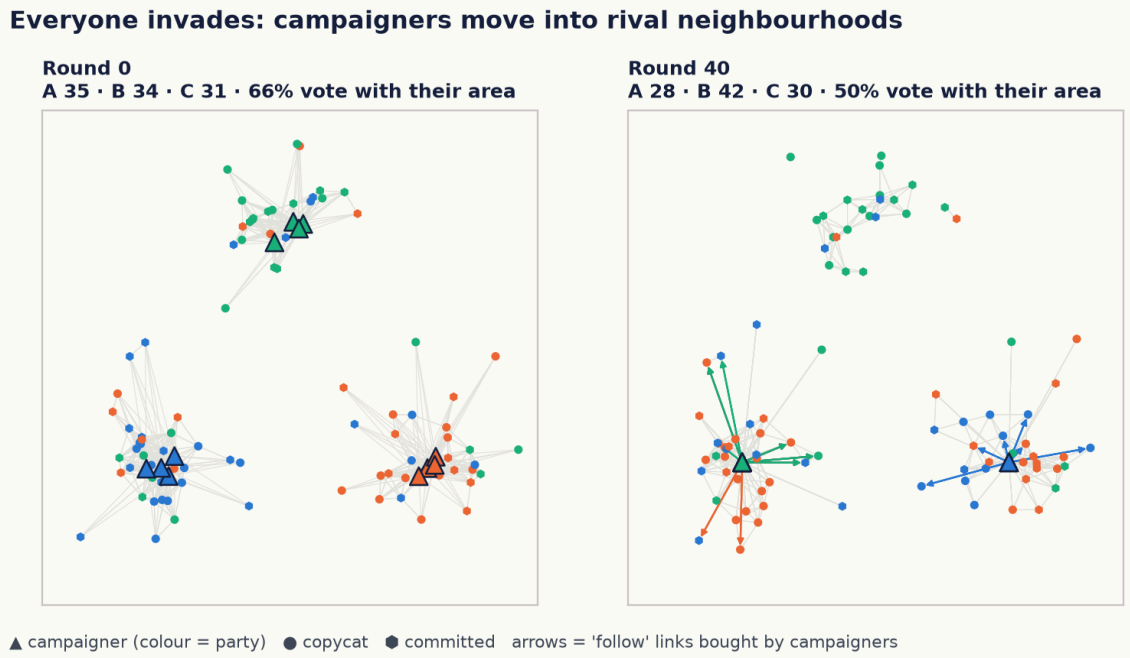
On a 41 × 41 lattice, long-run frequencies converge almost surely.

Pilot, 200 voters, 5 seeds

Share error, moving town: poll 0.024, blend 0.085, memory alone 0.167.

APPENDIX · EXTRA EXPERIMENT · SETUP

Campaigners who change who you hear



Everyone invades. Campaigners walk into rival neighbourhoods and get voters to follow them. By round 40 only 50% still vote with their neighbourhood.

THE TOWN

Three neighbourhoods, one per party: **A 34, B 30, C 24**. Most residents are copycats: they vote like the 4 or 5 neighbours they see, so each area becomes an **echo chamber**.

THE CAMPAIGNERS

4 per party. They move fast, can poll a wide area, and can get up to 6 voters to **follow** them, so those voters see the campaigner's vote.

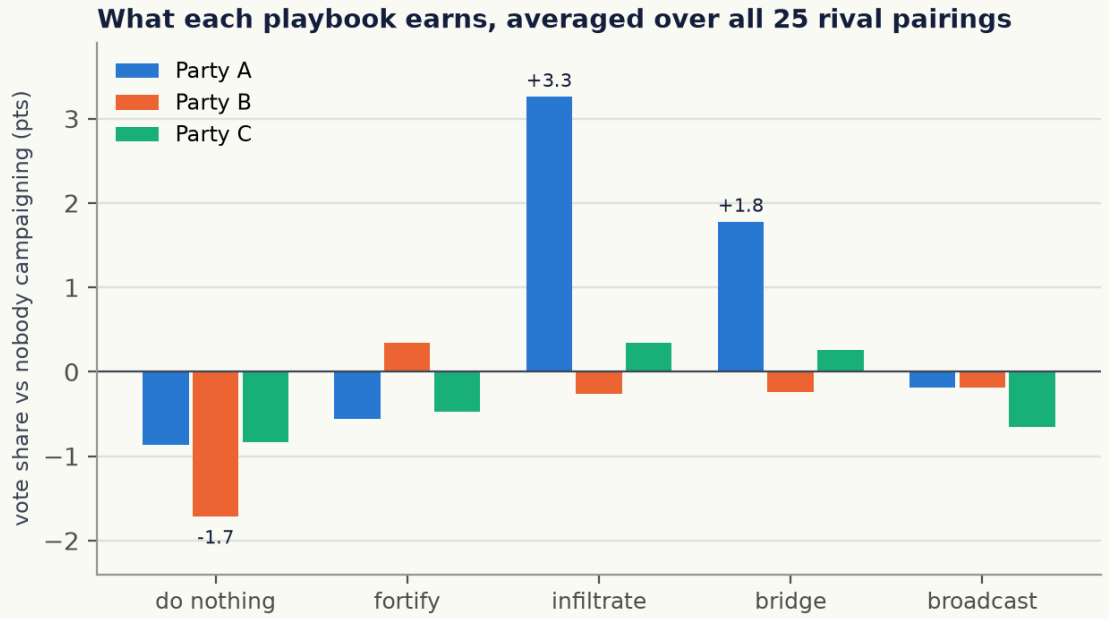
THE PLAYBOOKS

Do nothing · **defend** your home area · **invade** the biggest rival area · **work the border** · **random** followers. Every party picks one.

All 125 combinations of playbooks × 3 towns × 40 rounds · score: a party's vote share over the last 10 rounds

APPENDIX · EXTRA EXPERIMENT

Attack if you lead, defend if you chase



Each bar: how much a playbook earns a party on average, against every combination of its rivals' playbooks. “Fortify” is defend; “infiltrate” is invade.

WHAT WE GOT

A (biggest) should **invade**: +3.3 points. **B** should **defend**; doing nothing costs it 1.7. **C** (smallest) can't move the needle and won **0 of 375** elections.

DOES ANYONE SETTLE?

No. Each party keeps switching to beat the others, in a loop. The closest to stable is **everyone invading**.

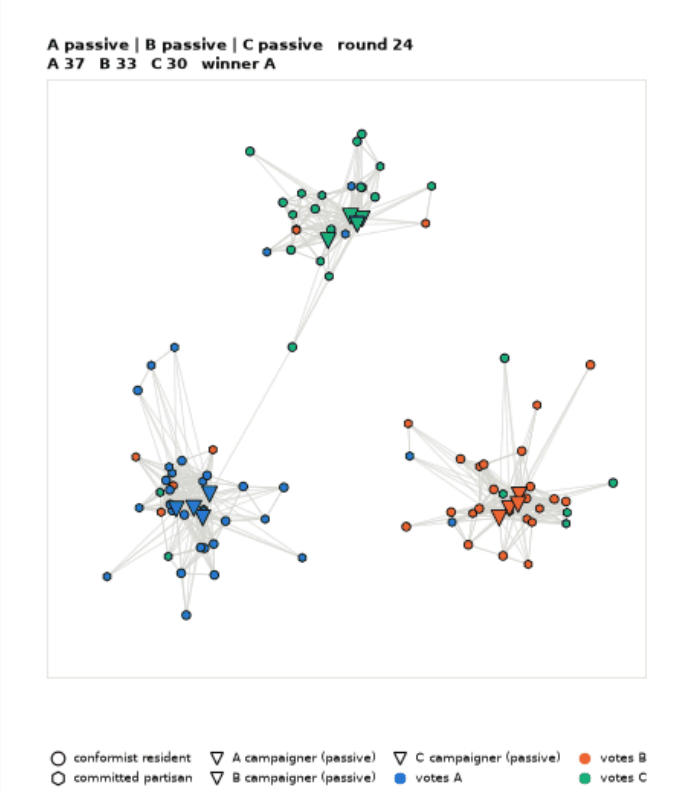
OUR REAL-WORLD READING

Campaign wars break echo chambers (79% → 55% vote with their area) but barely change who wins.

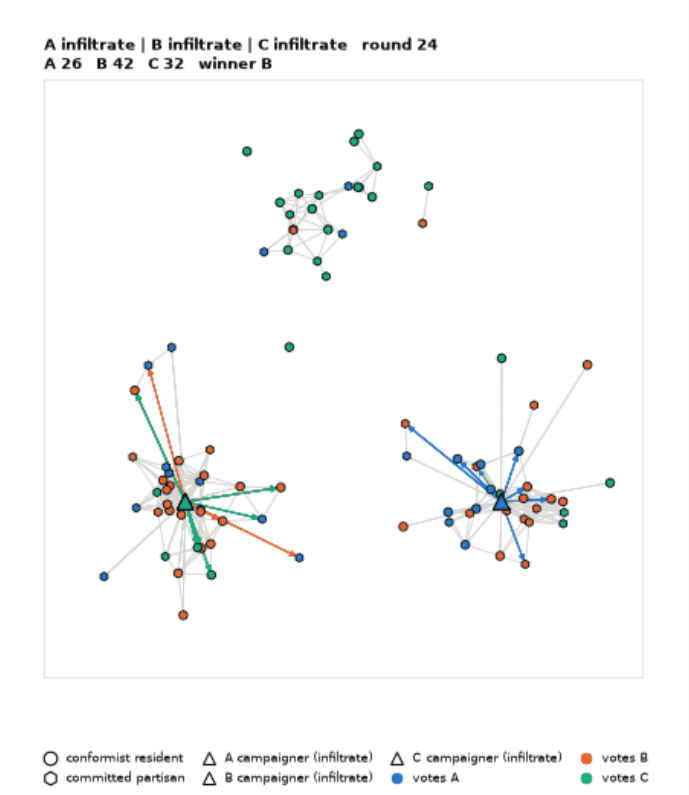
Across all 375 elections: A wins 348, B 27, C 0 · B's only upset route is working the border while A stays home

APPENDIX · EXTRA EXPERIMENT · WATCH IT HAPPEN

Nobody campaigns vs everyone invades



Nobody campaigns: neighbourhoods seal themselves and A wins.



Everyone invades: follow links tear A's area open and B takes the lead.

Town map · ▲ campaigners · ● copycats · ⬢ committed · arrows = follow links bought by campaigners